



N151AV (S/N: KONUK10295955) — Internal Board Interconnect (eth0) Hardware Fault Diagnostic Report

Immfly Ground Operations / IT — Session: September 8, 2026

Field	Value
Machine	N151AV (S/N: KONUK10295955)
Affected component	Internal Ethernet interconnect (eth0) between the OpenWRT/WiFi board and the main board
Initial symptom	Wireless network “aviancaonair.com” visible in client scan, but client unable to connect (stuck “obtaining IP address”)
Root cause	Confirmed — physical-layer fault in the eth0 cable/connector between the two internal boards. Link negotiates only at 100 Mbps and drops carrier intermittently, versus a stable 1000 Mbps link on the healthy reference unit (N127AV)
Impact	Wireless clients associate at Layer 2 successfully, but never receive a DHCP lease, because the only uplink path to the DHCP server on the main board (br-vlan150 port eth0.150) repeatedly loses carrier. WiFi service on this unit is effectively non-functional
Recommended action	Physical inspection, reseating, and replacement of the eth0 cable/connector between boards; if the fault persists after cable replacement, replace the affected board/NIC

Narrative Summary

Initial comparison of `/etc/config/dhcp`, `/etc/config/wireless`, and `/etc/config/network` between N151AV and the known-good reference unit N127AV found the wireless (AP, open encryption, SSID “aviancaonair.com”, VLAN 150) and network/bridge configuration to be identical. A stray **config dhcp 'lan'** stanza was found in N151AV's dhcp config (absent on N127AV) which was causing dnsmasq to interfere with DHCP broadcast traffic on br-vlan150 (“DHCP packet received on br-vlan150 which has no address” warnings). This stanza was removed via **uci delete dhcp.lan** and dnsmasq was restarted as a corrective step; the warning stopped and the wireless client associated cleanly, but the client still failed to obtain an IP address, ruling out configuration as the root cause.

Further diagnosis confirmed eth0.150 is correctly bridged into br-vlan150 alongside wlan0/wlan1. Interface counters on eth0.150 showed outbound traffic increasing during a client connection attempt while inbound traffic remained flat, indicating no return traffic from the main board. Kernel log analysis (dmesg) showed the physical eth0 link training only at phy_speed=100 and repeatedly reporting “GMAC Link is down” every few minutes throughout the session, always recovering at the same degraded speed. By contrast, the reference unit N127AV shows eth0 training and holding stable at phy_speed=1000. No CRC, RX, or TX errors were recorded on N151AV's eth0, indicating a clean, intermittent loss of carrier rather than signal noise — consistent with a damaged cable, a loose/oxidized connector, or a bent pin on the port, rather than EMI or a duplex mismatch. No cron jobs or userspace watchdog processes were

found that could account for the link cycling, ruling out a software-triggered cause.

Conclusion and Recommendation

N151AV's wireless connectivity failure is caused by a physical-layer fault in the eth0 interconnect between the OpenWRT/WiFi board and the main board, not by software or configuration. The dnsmasq configuration issue identified and corrected during this session was a secondary defect and has been resolved, but does not account for the connectivity failure. No further remote/software diagnostics are needed to confirm the hardware fault, as the degraded link speed and repeated carrier loss have already been captured in the kernel logs.

Recommended action: Escalate for physical inspection of the eth0 cable and connector between the two boards, following the standard repair-order workflow used for other confirmed hardware faults in the fleet. Reseat or replace the cable first and re-test with **dmesg | grep -iE "eth0.*GMAC Link"**, which should show a stable phy_speed=1000 with no further link-down events. If the fault persists after cable replacement, the Ethernet port/NIC on one of the two boards should be considered for replacement. As part of the repair, a full functional check (WiFi association, DHCP, and internet passthrough) is recommended before returning the unit to service.